Gold Price Prediction using Machine Learning

Phase 1 : Discovery Report

CN240 – Data Science for Signal Processing

Computer Engineering, Thammasat School of Engineering

Covering : Overview, Stakeholders, Framing the Problem, Objectives, Success Criteria, Formulating Hypotheses, Data Assessment, Resource Assessment & Roles, Project Risks and Conclusion

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Version 1.0 | March 2026

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1. Overview

Over the past decade, gold has evolved beyond its traditional role as a safe-haven asset into a complex investment instrument. The global economy during this period has been shaped by persistent uncertainty — from shifting monetary policies and the COVID-19 pandemic to intense geopolitical tensions such as U.S. tariff announcements and ongoing conflicts in the Middle East. These factors have driven gold prices in Thailand to levels of volatility where investment decisions can no longer rely on intuition alone.

This project applies concepts from CN240 – Data Science for Signal Processing to address the challenges of predicting prices in a highly volatile asset market. The year 2026 in particular has been marked by renewed instability from both financial policy changes and geopolitical developments, which have made historical price relationships increasingly difficult to interpret using conventional approaches.

To tackle this, our team built a full Data Science Pipeline — starting from Exploratory Data Analysis (EDA) and Feature Engineering, through to Model Comparison — with two main goals in mind: first, to develop a decision-support tool capable of distinguishing genuine investment signals from market noise; and second, to give every team member hands-on experience across the entire data science workflow.

2. Stakeholders

Since this project was developed for academic purposes — specifically to apply and test data science concepts — the stakeholders can be divided into two main groups.

2.1 Application-Oriented Stakeholders

The primary end-users of this project's findings are retail gold investors. Although the project is academic in nature, the trading signals and feature extraction techniques developed here could serve as a foundation for a basic decision-support tool. This would help individual investors reduce emotional bias and move toward a more data-driven approach when navigating volatile market conditions.

2.2 Academic and Development Stakeholders

The first group consists of the student team members themselves — computer engineering students who aim to apply data science theory to real-world time series data. The goal is to practice and validate end-to-end skills, covering everything from data preprocessing and feature engineering to model development and performance evaluation, across both regression and classification tasks.

The second group is the course instructor, who evaluates the project based on the correctness of the methodology, the reasoning behind model selection — particularly in handling non-stationary data — and the team's ability to interpret results in a way that addresses the research hypotheses.

3. Consider the Past

Studying past projects and related research is a crucial foundation for designing the model architecture for gold price prediction. Learning from previous quantitative tests helps our team avoid using the wrong algorithms and allows us to focus on solutions that have been proven to work in real-world scenarios. The key takeaways can be summarized as follows

3.1 Limitations of Traditional Models

Past studies show that using traditional statistical models and certain algorithms on financial time-series data often fails due to the following reasons:

- **Inflexibility to Non-linear Dynamics:** Traditional models like ARIMA or Naive Bayes usually struggle to handle large datasets or capture the complex relationships in financial markets [1, 2, 3] .
- **Weakness of Distance-Based Algorithms:** Models like K-Nearest Neighbors (KNN) [4] and Support Vector Machines (SVM) [5] are highly ineffective for predicting raw time-series data. Past evaluations showed that KNN had a high Mean Squared Error (MSE) of 61,218.36. Meanwhile, SVM struggled to adjust its hyperplane flexibly, resulting in a massive error of up to 407,690.66 [6].
- **Multicollinearity Issues:** Feeding highly correlated independent variables into a model makes regression coefficients unstable, which significantly reduces the model's accuracy [7].

3.2 Proven Architectures & Solutions

To overcome the limitations mentioned above, successful research often applies more advanced techniques:

- **Advanced Data Preprocessing:** Multicollinearity can be solved using Principal Component Analysis (PCA). This technique reduces the data's dimensions while keeping up to **96.37%** [8] of the original data's variance.
- **Effective Baseline Models:** Linear Regression and Ridge Regression serve as excellent baselines, achieving an MSE as low as 45.39 [6]. Additionally, Lasso Regression is outstanding for feature selection because it can reduce the coefficients of noisy variables to exactly zero [9].
- **The Power of Tree-Based Models:** Random Forest is highly robust against outliers and accurate in identifying trend directions[11]. Advanced algorithms like XGBoost and LightGBM are considered state-of-the-art because they minimize both bias and variance [10]. In one study, XGBoost achieved an Explained Variance Score higher than **0.959** and kept the Mean Absolute Error (MAE) at just 0.0238.
- **Deep Learning Networks:** Capturing long-term relationships requires models like LSTM [12], which achieved a low MSE of 376.603. Furthermore, Hybrid Frameworks (such as combining CNN with Bi-LSTM) improve predictions by blending feature extraction with forward and backward sequential analysis [13].

3.3 Thai Local Contexts

Since the primary objective of this study is to predict the physical gold price in Thailand (as benchmarked by the Gold Traders Association), the model cannot rely exclusively on the global spot gold price (XAU/USD). The pricing mechanism of domestic gold is inherently tied to the local exchange rate and importing costs. Mathematically, the Thai gold price formulation, as officially standardized by the Gold Traders Association and the Thailand Futures Exchange (TFEX) [14], [15], can be expressed as follows:

$$\mathbf{P_{TH}} = \left(\mathbf{P_{XAU}} + \mathbf{C_{prem}}\right) \times \mathbf{E_{FX}} \times \left(\frac{\mathbf{15.244}}{\mathbf{31.1035}}\right)$$

Where:

- $\mathbf{P_{TH}}$: represents the Thai physical gold price per Baht-weight (in THB).
- $\mathbf{P_{XAU}}$: is the global spot gold price (in USD per troy ounce).
- $\mathbf{C_{prem}}$: denotes the premium, including freight, insurance, and local importing costs.
- $\mathbf{E_{FX}}$: is the real-time USD/THB exchange rate.
- $\frac{\mathbf{15.244}}{\mathbf{31.1035}}$: is the standard weight conversion ratio from one troy ounce (31.1035 grams) to one Thai Baht-weight (15.244 grams).

According to this underlying mechanism, any machine learning architecture would fail to capture the true variance of the Thai gold market if the $\mathrm{E_{FX}}$ multiplier is omitted. Therefore, it is imperative to incorporate the USD/THB exchange rate and regional premium indicators as exogenous variables. Previous empirical research demonstrates that global gold prices can act as an effective exogenous variable, significantly improving the predictive accuracy of neural networks for foreign exchange rates—successfully reducing the RMSE to 0.0115. This finding confirms the deep, symbiotic relationship between these two assets, justifying their combined use as input features in our localized prediction framework.

4. Framing the Problem

Gold price volatility makes investment decisions inherently risky, as price movements are influenced by a complex mix of macroeconomic and geopolitical factors that are difficult to predict by intuition alone. To tackle this, the project applies machine learning techniques and frames the problem into two main parts:

1. **Regression** — predicting the closing price of gold on a given day.
2. **Classification** — identifying trading signals to categorize market conditions as buy, sell, or hold.

5. Objectives & Success Criteria

5.1 Project Objectives

- **To develop a gold price prediction model in volatile markets:** To build a Machine Learning model capable of handling gold market volatility. This includes predicting closing prices (Regression) and classifying market trends (Classification).
- **To extract and test alternative features:** To study the effectiveness of using proxy variables, such as the VIX Index and government bond yields, instead of Natural Language Processing (NLP) to capture safe-haven flows during crises.
- **To minimize false signals:** To apply time-series data techniques (Lag Features) and volatility indicators (like ATR) to filter out market noise and prevent fake breakouts.
- **To create an evidence-based investment strategy simulation:** To apply the fully developed and tuned models to backtesting. This will simulate financial returns and assess risks before real-world application.

5.2 Success Criteria

5.2.1 Dimension 1: Academic & Baseline Metrics

- **Baseline Performance:** A baseline model like Naive Bayes must predict market direction significantly better than random guessing (Accuracy $>$ **50%**).
- **Non-linear Model Accuracy:** Non-linear models (such as Decision Tree or Random Forest) must show lower error rates (RMSE or MAE) than a Linear Regression model when predicting closing prices.
- **Feature Effectiveness:** Adding proxy variables (VIX, Bond Yield) and volatility features (ATR) must improve the overall performance of the model compared to using only historical gold prices.

5.2.2 Dimension 2: Advanced Machine Learning Metrics

- **Precision (Signal Accuracy):** The Classification model must achieve a Precision score of **55% - 65%** for the "Buy" class. This confirms that the system prioritizes certainty. Missing an opportunity (False Negative) is acceptable and better than generating false buy signals (False Positive) that lead to financial losses.
- **ROC-AUC Score:** The model must be able to clearly distinguish between upward and downward market trends. The Area Under the Curve (ROC-AUC) must be at least **0.65 - 0.70**. This proves the model has real predictive power and is not just memorizing the training data (Overfitting).

5.2.3 Dimension 3: Financial & Risk Management Metrics

- **Net Profit:** After backtesting and deducting trading costs and fees, the simulated investment portfolio must show a positive total return.
- **Profit Factor:** The ratio of Gross Profit to Gross Loss within the system must be greater than **1.5**.

- **Maximum Drawdown (MDD):** The system must limit risks during severe market volatility. The maximum accumulated loss from a peak (MDD) must remain at an acceptable level (e.g., no more than **15-20%** of the portfolio). This proves the model is safe enough for practical use.

6. Formulating Hypotheses

6.1 Hypothesis 1 : Baseline Models

To establish a minimum performance lower bound, this project selects three baseline models from the curriculum, including Naive Bayes, Logistic Regression and Random Forest Classifier. Each model has different basic assumptions. Naive Bayes assumes that features are independent (Independence Assumption), which is known to contradict the highly correlated nature of price data; thus, it clearly serves as a statistical lower bound. Meanwhile, Logistic Regression has more flexible assumptions and is expected to perform better as a competitive baseline. All three models must be able to classify 'Buy' and 'Sell' signals significantly better than random guessing (Accuracy $> 50\%$) with statistical significance. This confirms that the selected data has enough predictive power before moving on to develop advanced models.

6.2 Hypothesis 2 : Time-Series and Volatility Feature Management

Creating and adding lag features (e.g., historical prices from the past 7-15 days) alongside volatility indicators (e.g., Average True Range [ATR] or Rolling Standard Deviation) will help the model distinguish real trend changes from temporary price fluctuations. This will effectively reduce the rate of false positive buy signals.

6.3 Hypothesis 3 : Macroeconomic Factors and Crisis Proxies

During periods of geopolitical market tension, using economic indicators like the "VIX Index" (Fear Index) and "US Government Bond Yields" as independent variables will act as effective proxies to capture safe-haven money flows. This can be achieved efficiently without needing a Natural Language Processing (NLP) system to analyze news articles.

6.4 Hypothesis 4 : Algorithm Complexity

Because gold prices and macroeconomic factors share a complex and non-linear relationship, using non-linear models (such as Decision Tree Regression or Random Forest) will predict gold closing prices (Regression) more accurately and result in lower error rates compared to simple Linear Regression models.

7. Data Assessment

7.1 Engineered Features

The table below details the variables that have completed the feature engineering process. These will be used as the actual inputs for training the model. These variables are transformed from the raw data mentioned in Section 7.2.

Feature	Data Type	Unit	Frequency
XAU/USD Log Returns	Numeric (Float)	Decimal Proportion	Daily
30-Day Realized Volatility	Numeric (Float)	Percentage (Annualized)	Daily (Rolling)
VIX 30-Day Z-Score	Numeric (Float)	Standard Deviation	Daily (Rolling)
USD/THB Log Returns	Numeric (Float)	Decimal Proportion	Daily

7.2 Data Scope & Sourcing

This section explains the sources and scope of the raw data before processing. The raw dataset contains more variables than those shown in Table 7.1. Some variables, such as Brent Crude Oil, WTI, and GVZ, were considered during the initial screening but did not meet the correlation criteria. Therefore, they were excluded from the final engineered features

- **Timeframe:** The data covers the period from 2010 to the end of 2025 (01/01/2010 – 31/12/2025).
- **Target Variable:** The price of 96.5% purity gold bullion, sourced from the Gold Traders Association (GTA).
- **Independent Variables (Predictors):**
  - **Global Gold Spot (XAU/USD):** The global gold price, which serves as the primary benchmark.
  - **Currency Exchange Rate (USD/THB):** The exchange rate between the US Dollar and the Thai Baht, representing the translation effect.
  - **Brent Crude Oil and WTI:** Oil prices, which are correlated with inflation rates.
  - **Market Sentiment & Volatility Indices:**
    - **CBOE Volatility Index (VIX):** The "fear index" of the US stock market.
    - **CBOE Gold Volatility Index (GVZ):** The volatility index specifically for gold prices.

7.3 Data Quality & Pre-processing

7.3.1 Datetime Formatting & Indexing

- **Problem:** The raw data from different files had inconsistent date formats and were stored as plain text.
- **Cleaning Process:** We converted these text values into a standard Datetime format. We then set the time column as the table's index and sorted it chronologically (from past to present). This sorting step is critical; without it, all time-series calculations would be distorted.

7.3.2 Data Merging & Alignment

- **Problem:** The gold market, the currency exchange market, and the US stock market (VIX) have different operating hours and public holidays. This resulted in an unequal number of data rows and missing dates.
- **Cleaning Process:** We merged the datasets using an "Inner Join" method based on the Datetime index. This automatically filters the data, keeping only the dates where all three markets have complete information. Any row with missing data from any market (due to holidays) was safely dropped.

7.3.3 Feature Engineering

Because raw prices are not stationary (as proven by the ADF Test), we cannot feed them directly into the model. We transformed the data as follows:

- **XAU/USD & USD/THB Log Return:** We used the natural logarithm equation **ln(Current Price / Yesterday's Price)** to convert raw prices into daily percentage fluctuations.
- **VIX 30D Z-Score:** We created a 30-day rolling window to find the mean and standard deviation (SD). We then subtracted the mean from the current VIX value and divided it by the SD. This measures how "abnormal" current market fear is compared to the recent average.
- **Global Gold 30D Rolling Realized Volatility:** We used the XAU Log Returns to calculate the 30-day historical volatility (SD). We then annualized this value by multiplying it by the square root of 252 (the number of trading days in a year) to measure the intensity of price swings.

7.3.4 Dropping NaN Values

- **Problem:** When applying backward-looking calculations (like 30-day rolling windows and 1-day shifts), the first 30 rows of the dataset become empty (NaN) because there is no past data to calculate from.
- **Cleaning Process:** We used the **.dropna()** command to completely remove these first 30 invalid rows. If even a single NaN (Not a Number) value remains, AI algorithms and statistical models will immediately fail. To compensate for this dropped data, we can increase the volume of historical data collected.

7.3.5 Train/Test Split Strategy

Because this project uses time-series data, we must strictly avoid random splitting. Random splitting causes "Data Leakage," where future data leaks into the training set, resulting in unrealistically high model performance.

- The team will primarily use a **Time-Based Split** strategy. Data from 2010 to 2022 will serve as the Training Set, while data from 2023 to 2025 will be the Test Set. This simulates a real-world scenario where the model must predict unseen future data.
- Additionally, during Hyperparameter Tuning, we will consider using **Walk-Forward Validation**. This technique shifts the data window forward over time to evaluate the model's stability across different market cycles, such as the COVID-19 crisis and periods of geopolitical tension.

7.4 Initial Correlation Assessment

- **Strong Positive Correlation:** When evaluating the relationship of these factors using scatter plots grouped by market direction, Thai gold bullion prices show a strong positive correlation with global gold returns (XAU/USD) and the exchange rate (USD/THB). The data clusters neatly according to upward and downward trends. This indicates that when global gold prices rise and the Thai Baht depreciates, domestic gold prices increase, which perfectly aligns with economic principles.
- **Volatility Indicators:** While volatility variables (VIX and Realized Volatility) do not display a clear linear relationship, the Kernel Density Estimation (KDE) plots reveal that during periods of high volatility, the data points spread much wider. This visually represents the "size" of the investment risk.
- **Low Correlation Factors:** Conversely, Brent Crude Oil and WTI show a low correlation. The mechanism that transfers the impact of energy prices to safe-haven assets requires time to reflect in inflation indices (Lagged Effect). This delayed effect does not match the behavior of daily price movements.
- **Conclusion:** Based on the first round of evaluation, we selected the three factors most correlated with Thai gold prices: **XAU/USD, USD/THB and VIX**. These variables possess structural and statistical readiness, demonstrate stationarity, and comprehensively cover different dimensions of risk. They will serve as the crucial foundation for building an accurate and reliable gold price forecasting model for highly volatile markets.

8. Resource Assessment & Roles

To execute the project effectively, we have assessed our resource readiness and defined an initial team structure to ensure systematic and efficient management. The details are as follows:

8.1 Technological Resources

The infrastructure and tools for the initial phase focus on agility and team collaboration. These include:

- **Environment & Compute:** We use Google Colab as our primary coding environment because it supports real-time collaboration and reduces personal hardware limitations.
- **Version Control:** We use GitHub to store code, control development versions, and allow members to integrate their work systematically.
- **Project Management:** We use GitHub's Kanban board to track task progress. This makes it easy to manage sub-tasks in each phase and effectively monitor the team's workflow.

8.2 Data Resources

The project uses data covering the period from 2010 to the end of 2025 (January 1, 2010, to December 31, 2025). The data sources are as follows:

- **Domestic Gold Bullion Prices:** Sourced from the Gold Traders Association (GTA) website (https://www.goldtraders.or.th).
- **Global Gold Spot (XAU/USD):** Sourced from MetaTrader 5.
- **Currency Exchange Rate (USD/THB):** Sourced from MetaTrader 5.
- **Brent Crude Oil and WTI:** Sourced from http://investing.com.
- **VIX and GVZ:** Sourced from https://www.cboe.com/tradable-products/vix/vix-historical-data.
- **Supplementary Datasets:** Where applicable, the Hugging Face Datasets library is used to access and integrate publicly available financial or economic datasets that may support model training and benchmarking.

8.3 Time Resources

- **Expected Delivery Date:** May 12, 2026

8.4 Human Resources & Roles

To ensure clear operations in Phase 1, core responsibilities have been assigned to team members. *(Note: A single member can hold multiple roles, and some roles can be shared among members depending on the workflow.)*

The role assignments are as follows:

- **Project Manager:** Athiphat Sunsit
    - **Main Duties:** Acts as the center for planning and managing the team's daily tasks. Organizes and tracks progress via the Kanban Board, ensures everyone meets milestones, and coordinates or solves immediate problems to keep the project moving forward.
- **Business Intelligence:** Napattheera Laokoamhung
    - **Main Duties:** Understands project goals, analyzes success metrics, and determines the presentation format that best meets the practical needs of the project.
- **Data Engineer:** Theepob Rattanapsapsiri
    - **Main Duties:** Plans the data pipeline structure, evaluates data sources, and prepares data extraction into the analysis system.
- **Database Administrator:** Patcharapon Malaisri
    - **Main Duties:** Evaluates and designs the data storage architecture to ensure it can fully support the datasets used in the project.
- **Data Scientists:** Purich Ampawa and Panithan Thantue
    - **Main Duties:** Work together to frame problems, formulate hypotheses, assess data relevance, and plan the analytical techniques or models for the next phase.

Beyond their main responsibilities, every team member supports all aspects of the project. The primary goal is for everyone to learn. This approach provides an opportunity for all members to rotate and experience every position, ensuring that everyone fully understands how to develop a Data Science project using Machine Learning according to the project's objectives.

9. Project Risks

In developing the gold price forecasting model for the Gold ML Prediction project, we have evaluated three main risks that could impact the model's performance:

9.1 Risk from Not Using Natural Language Processing (NLP) for News Analysis

- **Cause:** The project uses the VIX index and government bond yields as proxies to capture money flows, rather than using NLP techniques to directly analyze the context of news articles.
- **Impact:** The model may respond slowly to sudden emergencies (such as sudden declarations of war or new political policies). It must wait for market panic to reflect in the VIX or GVZ indices first, which could create a disadvantage in real-time decision-making.
- **Mitigation:** The team will focus on using models that effectively capture volatility and evaluate the impacts reflected in lag features. This ensures the model can respond to changing statistical trends as quickly as possible.

9.2 Risk from Unprecedented Historical Events (Black Swan Events)

- **Cause:** The model is trained using historical data from 2010 to the end of 2025.
- **Impact:** If a new type of economic crisis or unprecedented geopolitical instability occurs in 2026, the historical relationships between the analyzed variables (such as XAU/USD and volatility) may change. This could lead to a decrease in the model's accuracy.
- **Mitigation:** The system is designed to continuously perform backtesting and closely monitor the Maximum Drawdown (MDD). If the simulated portfolio experiences losses exceeding an acceptable threshold, it will serve as a clear signal for the team to perform Model Retraining.

9.3 Risk from Differences in Market Trading Hours

- **Cause:** Global gold prices and reference indices are traded almost 24 hours a day, whereas domestic gold bullion prices follow standard Thai market opening and closing hours.
- **Impact:** Even though we use an "Inner Join" technique to resolve mismatched market holidays, price volatility that occurs outside of Thai trading hours (such as overnight) may not be perfectly reflected in the Thai closing price.
- **Mitigation:** Converting the raw price data into Daily Logarithmic Returns will help adjust for data non-stationarity and, to some extent, reduce the impact of sudden price jumps when the market opens.

10. Conclusion

The initial phase of development focused on **framing the problem** and establishing a robust foundation for the machine learning models designed to analyze and forecast highly volatile gold prices. This stage involved defining the core objectives, success criteria, and research hypotheses. The team assessed and selected critical datasets, including **Global Spot Gold (XAU/USD)**, the **USD/THB exchange rate**, and **market volatility indices (VIX)**, to serve as key features for model construction.

Furthermore, comprehensive **data quality assessments** were conducted. Data was pre-processed to suit time-series analysis, utilizing techniques such as **Log Returns transformation** and **Stationarity testing (Unit Root tests)**. Preliminary correlation analysis indicates that specific exogenous variables maintain a statistically significant relationship with Thai gold prices, reinforcing the project's initial hypotheses.

The report also details the assessment of technological resources, team structure, and potential project risks. This foundational work in problem definition and data preparation sets a clear trajectory for the upcoming development and modeling phases.

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